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SignAlert: An Agentic Vision Framework for Real-Time Detection and Escalation of Medical Distress Signs in Indian Sign Language

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PROJECT REVIEW I — PROBLEM DEFINITION, DESIGN AND PLANNING REPORT

1. Problem Statement

Deaf and non-verbal individuals living alone or with medical vulnerabilities — cardiac conditions, seizure disorders, or elderly relatives living unsupervised — currently have no fast, independent way to signal a medical emergency. Voice-based options such as smart speakers, calling emergency numbers, or shouting for help are all built for a hearing, speaking user and are simply not accessible to this population. A person who collapses, feels dizzy, or begins showing symptoms of a heart attack has no channel to alert a caregiver except being physically seen by someone who understands sign language, at the exact moment it happens.

SignAlert closes this gap by watching a monitored space continuously, recognising a small, high-value vocabulary of Indian Sign Language (ISL) distress signs — heart attack, breathlessness, dizziness, pain, and a general call for help — directly from a camera feed, and then acting on what it sees rather than simply logging it. It also does not depend on the person being able to sign at all: a fall or an unusual period of stillness is detected the same way, so a person who has lost consciousness is still protected.

Existing solutions stop short of this. Fall-detection wearables exist but are not sign-language aware and require the person to already be wearing a device. General-purpose home security cameras can alert someone that motion occurred, but cannot distinguish a medical emergency from ordinary movement, and none of them carry a request through to a human response. SignAlert's contribution is the closed loop: detect the distress signal, confirm it is not a false alarm, and autonomously work through caregivers, family, and — if nobody responds — emergency services, without requiring the person to speak, dial, or type.

The scale of this gap is easy to underestimate precisely because it is invisible in most safety statistics: a Deaf or non-verbal person who is never able to report their own emergency does not show up as a failed emergency call, they simply show up late, or not at all, to a response that could have started minutes earlier. Framing the problem this way — as a communication-channel gap rather than a general home-safety gap — is what distinguishes SignAlert's scope from a generic monitoring or wearable product, and is the analytical objective this project is built around: close the specific gap between a Deaf or non-verbal person's native channel of expression and the automated systems meant to protect them.

2. Business Use Case

Emergencies involving Deaf and non-verbal individuals living alone follow a recurring pattern: the person has a way to express what is wrong through their sign language, but no one is present who understands it, and no automated system is watching for it. By the time a caregiver happens to check in, the window to respond safely may already have closed. Beyond the immediate danger to the individual, families and care facilities carry lasting anxiety and liability from not knowing whether a loved one is safe when unsupervised.

SignAlert's value lies in what it does not require: no change to how the person communicates, no wearable they have to remember to charge, and no new hardware beyond a camera already common in home-monitoring setups. It layers directly onto a standard camera feed and does one job well — compressing the time between something going wrong and a caregiver knowing about it, from indefinite (until someone happens to check) to under a minute, without needing the person to press a button or speak.

Just as important is how little this system takes away from the person's autonomy: no video ever needs to leave the room unless a genuine distress event is confirmed, and the person can cancel a false alarm with a single gesture before any notification goes out. This makes it realistic to trial in a single household or care room without asking the person to accept blanket surveillance in exchange for safety — for the cost of a camera and a software layer, SignAlert turns an unmonitored, reactive situation into a proactive one.

From a business standpoint, the cost of inaction is asymmetric: a missed or delayed emergency response carries a severity that no amount of convenience elsewhere in a care setting can offset, while the cost of deploying SignAlert is limited to a camera the household or facility likely already owns and a software layer that runs on modest hardware. This asymmetry is what makes a narrow, safety-critical vocabulary the right starting scope rather than a limitation — the project is deliberately optimised to be right and fast on a small set of high-stakes signals before it is asked to be broad.

3. User and Stakeholder Analysis

Five distinct stakeholder groups interact with SignAlert, each with a different relationship to the system and a different pain point that the current absence of any sign-language-aware monitoring leaves unaddressed. These are summarised below.

Stakeholder	Role / Interest	Needs from SignAlert	Current Pain Point
Deaf / Non-verbal Individual	The person being protected; communicates in ISL	A system that understands their sign language, not just generic motion or sound	No existing system recognises sign-language distress cues at all
Primary Caregiver / Relative	First point of contact when something goes wrong	An immediate, specific alert (what happened, where) instead of finding out too late	Currently only finds out by chance, or after the fact
Secondary Contacts / Family Group	Backup responders if the primary caregiver is unreachable	To be looped in automatically, not manually, if the first stage fails	No structured escalation path exists today
Emergency Services	Final responder if no human contact answers in time	A clear, text-based description of the emergency and location, since the reporter cannot call or speak	No accessible channel for a Deaf or non-verbal person to report their own emergency
Care Facility / Institution	Owns responsibility for resident safety and liability	An auditable log of what was detected, who was notified, and when	No structured record of real-time monitoring or response today

4. Data Source and Knowledge Source Identification

The Perception agent is built on MediaPipe Holistic, a pretrained pose, face, and hand landmark model. It is not trained from scratch — an existing, well-validated checkpoint is used — and it converts each video frame into a compact set of body and hand keypoints. These keypoints are fed into a temporal sequence classifier (LSTM/Transformer) trained specifically on a curated ISL distress vocabulary — heart attack, breathlessness, dizziness, pain, help, and a manual panic gesture — rather than attempting full sign-language recognition, which is a far larger and less reliable undertaking for a project of this scope.

Landmark-based recognition was chosen over raw pixel classification because it holds up far better across different body types, clothing, lighting, and camera angles, and because it is dramatically cheaper to train a temporal classifier on a small vocabulary of keypoint sequences than to train a full video classifier from pixels. No public dataset combines ISL distress vocabulary with labelled medical-emergency video, so this project relies on a small, purpose-recorded dataset built with willing ISL-fluent participants, supplemented by publicly available general ISL sign datasets — such as INCLUDE — for pretraining the sign-recognition backbone before fine-tuning on the distress-specific vocabulary.

Landmark data alone cannot distinguish a real medical emergency from someone practising the sign for fun, or from an unrelated but similar-looking gesture. This gap is filled by the Confirmation layer, which requires the sign to be held or repeated across multiple consecutive frames above a confidence threshold, plus a short buffer window in which the person can cancel with a simple gesture before anything is escalated. Fall and stillness detection use pose-based heuristics — a sudden vertical drop, or a prolonged absence of expected movement — rather than a separate learned model, since these patterns are well-characterised geometrically and do not require training data of their own.

5. Data Understanding and Preprocessing Profile

Before any model is trained, every recorded clip is profiled to confirm it is fit for use: clip duration (nominally two to four seconds), frame count, and landmark-detection coverage are checked so that clips where MediaPipe repeatedly fails to detect a hand or body part — for example, due to poor lighting or the signer stepping partly out of frame — can be flagged and re-recorded rather than silently degrading the training set. Because MediaPipe fills in zero-valued landmarks whenever a hand or body part is not visible in a given frame, the pipeline degrades gracefully rather than failing outright, but a clip with an unusually high proportion of zero-valued frames is still treated as a data-quality signal worth reviewing rather than accepted automatically.

Each frame contributes a 258-dimensional feature vector: 33 pose landmarks with x, y, z, and visibility values, plus 21 left-hand and 21 right-hand landmarks with x, y, z values. To normalise clips of varying length into a fixed input shape, 30 frames are sampled per clip using linear interpolation over frame indices, producing a uniform (30×258) tensor for every example regardless of whether the source recording ran two or four seconds. Class balance across the five distress signs is checked at this stage as well, and the 80/20 train-test split is stratified specifically to preserve that balance in both partitions,

since an uneven split on a small, purpose-recorded dataset could otherwise make the reported evaluation numbers misleading.

This profiling step is treated as a first-class part of the pipeline rather than a one-off check, because the dataset is small and self-collected: a handful of mislabelled or low-quality clips would have a proportionally much larger effect on a dataset of this size than on a large public benchmark. Reviewing landmark coverage, class balance, and clip-length distribution before training is what gives the team confidence that the model's reported accuracy reflects genuine signal in the landmark data rather than an artefact of a skewed or noisy training set.

6. Selection of Tools and Frameworks

Each architectural layer was matched to the tool best suited to its specific constraints, favouring approaches that remain realistic to build and demonstrate at project scale without requiring a large paid infrastructure footprint. For perception and computer vision, MediaPipe Holistic landmarks feeding an LSTM/Transformer sequence classifier were chosen because landmark-based recognition generalises far better across users and lighting than raw pixel classification, and is realistic to train on a small distress vocabulary without a large dataset. Fallback detection uses pose-based heuristics for falls and stillness rather than a learned model, since these patterns are well-characterised geometrically, making a lightweight rule-based detector both more reliable and more explainable than a model trained on limited fall data.

Agent orchestration is handled by LangGraph, which is purpose-built for a state machine with a confirmation interrupt and a multi-stage escalation and feedback loop — a closer structural fit than a general-purpose multi-agent chat framework such as CrewAI or AutoGen. Where judgment is genuinely needed — turning a detected sign and its surrounding context into a clear text description for caregivers and emergency services — the reasoning layer uses the DeepSeek API, or a locally hosted Qwen3 model served through Ollama, falling back to a fixed template if no API key is set so that the demo still runs offline.

On the infrastructure side, the backend is built on FastAPI for its async-friendly design and native WebSocket support for live camera status, with the added benefit of auto-generated Swagger documentation for easy demonstration. The caregiver-facing application is a single-file HTML, CSS, and JavaScript dashboard with a WebSocket client, requiring no build step while still driving live status and alert history without additional infrastructure. Notifications are delivered through Twilio's SMS and push notification API, which falls back to a clearly labelled simulated SMS or push log if no credentials are configured, so the demo works end-to-end even without a paid account.

7. Agent Workflow and Orchestration Design

SignAlert is organised as seven cooperating agents, coordinated by a single orchestrator rather than left to communicate freely. The Perception agent converts each incoming video frame into structured sign and pose data, producing a sign identifier, a confidence score, a pose state, and a timestamp. Running alongside it, the Fallback Detector watches for falls or prolonged stillness independent of

whether the person is signing at all, producing an event type — fall or stillness — together with its own confidence score.

Neither of these raw detections is escalated directly. The Confirmation agent filters false positives first: a detection is only marked confirmed after it survives a short buffer window and is not cancelled by the person's own gesture. Once confirmed, the Risk Assessment agent reasons about severity using the detected sign and surrounding context — whether the sign has been repeated, whether a panic gesture has been given, or whether an earlier prompt went unanswered — and recommends a severity level and an escalation stage rather than escalating on the strength of a single detection alone.

The Escalation Controller then drives the fixed five-stage escalation ladder: a local, on-device alert; a notification to the primary caregiver; a notification to secondary contacts if the caregiver does not respond; escalation toward emergency services if no human contact responds; and continuous event logging at every stage. Each stage transition advances only on an explicit timeout or an immediate override triggered by a repeated, high-confidence panic gesture. The Dispatch agent is responsible for actually delivering the notification or text-based emergency description associated with the current stage, and location where available.

All of this is coordinated by a single Orchestrator, implemented as one LangGraph state-machine instance per monitored room, which routes state between agents and enforces the confirmation checkpoint before any escalation stage can advance. This state-machine-oriented design was chosen deliberately over a general-purpose multi-agent chat framework, since the escalation workflow has explicit stages and conditional branching rather than open-ended dialogue between agents — a structural fit that keeps the system's behaviour predictable and auditable, which matters given that its output is a real-world emergency notification rather than a conversational response.

End to end, a single distress event therefore moves through a predictable data flow: a video frame arrives at Perception and Fallback Detection in parallel; whichever agent produces a positive detection hands it to Confirmation; a confirmed event is scored by Risk Assessment; the resulting severity and recommended stage are handed to the Escalation Controller, which advances the ladder on its own timeout logic; and Dispatch is invoked at each stage to actually deliver a notification. Every step in this chain writes to the room's event log, so the full sequence of what was detected, confirmed, assessed, and escalated can be reconstructed after the fact — a property that matters as much for debugging during development as it will for accountability once the system is deployed.

8. Success Metrics and Evaluation Criteria

Because SignAlert's output is a real-world action rather than a label, success is defined across two layers rather than one: how accurately the perception model recognises the trained vocabulary, and how reliably the agentic pipeline converts a genuine detection into a completed notification. Optimising only the first would miss the point of the project; optimising only the second is meaningless without an accurate detector underneath it. The metrics below were defined at the start of the project specifically so that progress could be measured consistently across both layers as the system moves from prototype into a more complete build.

Metric	Target	Rationale
Sign-recognition test accuracy	≥ 90% on the 5-class vocabulary	Sets a bar high enough that confirmation and escalation logic are not compensating for a weak classifier.
False-positive rate after confirmation	As low as practically achievable	A caregiver who receives frequent false alarms will start to distrust and ignore real ones.
Confirmation-to-first-notification latency	Under the 5–10 s confirmation window plus dispatch time	The system's core value proposition is speed; a slow first notification undermines the whole design.
Escalation continuity under disconnection	100% of in-flight escalations complete on the backend	An emergency response must not be silently cancelled by a lost camera or client connection.
End-to-end demo completion	Full camera-to-SMS flow runs without manual intervention	Confirms the agents are genuinely orchestrated, not manually chained together for the demo.

These targets are intentionally modest at this stage rather than aspirational, since the project's first priority is demonstrating that the closed loop — detection through to a delivered notification — works reliably end to end on a small vocabulary. Raising the accuracy bar, tightening the false-positive rate, and validating the latency and continuity targets against real usage rather than a scripted demo are treated as the natural next stage of evaluation once the functional prototype is in place.

9. Risks and Mitigations

The most direct risk to the project is a false negative — a genuine distress event that the system fails to detect or fails to escalate. This is mitigated by design rather than by hoping the model is accurate enough on its own: the fallback layer means a person does not need to successfully sign at all to be protected, since a fall or prolonged stillness is caught independently, and the escalation ladder's timeout-driven structure means that even an ambiguous or low-confidence event is not simply dropped — it can still advance if it recurs or is reinforced by a panic gesture.

A closely related risk runs in the opposite direction: false positives that erode trust in the system faster than they cause direct harm. Frequent unnecessary alerts would train caregivers to deprioritise notifications from SignAlert, which is arguably as dangerous as a missed detection, since it undermines the very response the system is meant to trigger. The confirmation layer's multi-frame thresholding and cancel-gesture window exist specifically to keep this rate low without weakening the underlying safety net.

A third risk is more practical than technical: dependency on external services that may not always be available. The Twilio integration is currently limited by trial-account restrictions to pre-verified numbers, and the reasoning layer depends on an external LLM API or a locally hosted model. Both dependencies are mitigated by deterministic fallbacks — a simulated notification log and a fixed message template respectively — so that neither service being unavailable prevents the core escalation

logic from running and being demonstrated. Finally, the small size of the purpose-recorded dataset is a data risk in its own right: it is mitigated in the near term by careful data-quality profiling (see Section 5) and is flagged explicitly as an area requiring expansion before any real-world deployment is considered.

10. Ethical and Privacy Considerations

A system that continuously observes a person in their home, even for their own protection, raises legitimate privacy concerns that need to be treated as design requirements rather than an afterthought. The architecture assumes that landmark extraction and inference happen on-device or on a local server, so raw video does not need to leave the monitored space during normal operation; only a text description of a confirmed event, not video, is sent onward to caregivers or emergency services.

Deployment of a system like this should be accompanied by explicit, informed consent from the monitored individual, a visible and easily accessible way to disable monitoring, and a clearly stated data retention policy — all of which are treated as first-class requirements in the broader project this report is drawn from, not optional extras to be added later. A system's technical accuracy does not, by itself, justify its deployment without corresponding attention to the autonomy and dignity of the person it is intended to protect, and the team has kept this principle central to every design decision described in this report, from the on-device inference assumption to the single-gesture cancel option available to the person being monitored.

11. Project Plan and Milestones

The project has progressed through nine planned milestones so far, each building directly on the outcome of the one before it. Work began by grounding the problem: identifying the lack of any sign-language-aware emergency detection system for Deaf and non-verbal individuals living alone, which made the case for an agentic solution evidence-based rather than assumed. This was followed by designing the agent architecture itself — defining the perception, fallback, confirmation, risk assessment, escalation, and dispatch agent split, with a fixed five-stage escalation ladder as the core design commitment.

With the architecture defined, the team selected a small, high-value ISL distress vocabulary — heart attack, breathlessness, dizziness, pain, help, and a panic override — and identified INCLUDE and purpose-recorded footage as the data sources to train against. The technology stack was chosen next: MediaPipe Holistic with an LSTM/Transformer classifier for perception, LangGraph for orchestration, and confirmed fallback options for reasoning and notifications, so that the demo works reliably even without paid API keys.

To de-risk integration work before a live camera and real participants were available, a synthetic event simulator was built that replays labelled distress-sign and fall sequences, letting every agent be tested independently during early development. This was followed by integrating a working implementation — connecting the FastAPI backend, the LangGraph orchestrator, and the MediaPipe perception pipeline into a single end-to-end flow from camera frame to escalation stage — and then adding the

confirmation layer itself, implementing the buffer-window cancel gesture and multi-frame confidence thresholding to reduce false positives before any notification is sent.

Notifications and the caregiver-facing application followed: caregiver contact configuration and Twilio integration, with a simulated fallback, were wired into the Dispatch agent, alongside a live status dashboard for caregivers. The most recent milestone expanded the demo's requirements further, adding an admin setup flow for room configuration, caregiver contact lists, and escalation timeouts, together with a redesigned dashboard showing live camera status, alert history, and the current escalation stage — the state the project is in as this review is submitted.

12. Conclusion

SignAlert reframes emergency response for Deaf and non-verbal individuals around how they already communicate, rather than asking them to adapt to a system built for hearing, speaking users. By recognising a focused ISL distress vocabulary alongside fall and stillness detection, the system does not depend on the person being able to sign in the moment — it protects them whether they are able to gesture for help or have already lost consciousness. The confirmation layer and manual override keep false alarms low without weakening the safety net, and the fixed five-stage escalation ladder ensures that a detected emergency is never left unanswered: it moves from local alert to caregiver to family to emergency services until a human responds.

Just as important is what the system does not require: no new hardware beyond a camera, no wearable to remember, and no change to how the person expresses themselves. On-device inference and explicit consent keep the person in control of their own privacy, so protection does not come at the cost of dignity. Built on this foundation, SignAlert demonstrates that an agentic system can move beyond passive monitoring into genuine, accountable action — closing the gap between a person signalling distress and a caregiver actually knowing about it.